

Unit 11, Lesson 7: Translating Real-World Written Descriptions into Algebraic Inequalities

Benchmark

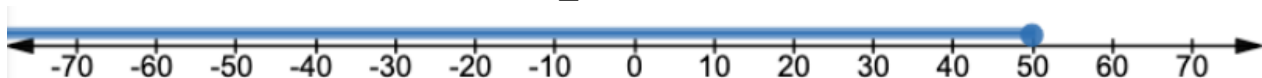
MA.6.AR.1.2 Translate a real-world written description into an algebraic inequality in the form of $x > a$, $x < a$, $x \geq a$, or $x \leq a$. Represent the inequality on a number line.

Learning Targets

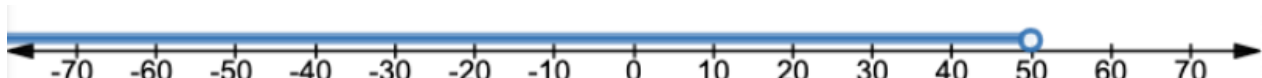
- ☐ I can translate a written description into an algebraic inequality.
- ☐ I can represent an inequality on a number line when given a written description of a real-world context.

11.7.2 Exploration: Representing Inequalities on a Number Line

1. Complete the following problems.
 - a. Choose an x -value that will make the inequality $x \leq 50$ true.
 - b. Choose an x -value that will make the inequality $x \leq 50$ false.
 - c. Have your partner verify your choices are correct.
 - d. What is the x -value that is true for $x \leq 50$ but false for $x < 50$?
 - e. The number line below represents all of the x -values that are true for $x \leq 50$.



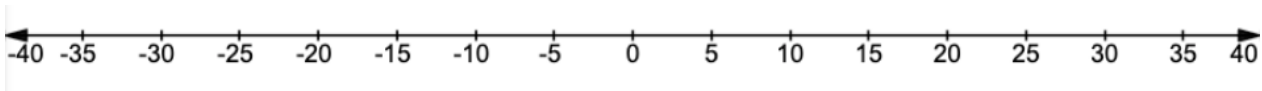
The number line below represents all of the x -values that are true for $x < 50$.



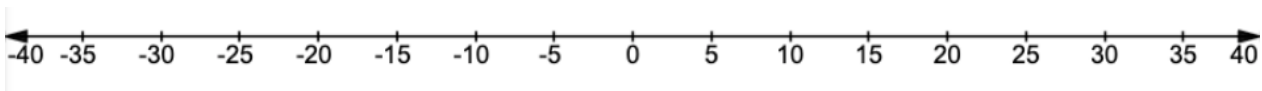
How is the graph for $x < 50$ different from the graph for $x \leq 50$?

2. Complete the following.
- a. How are the solutions to $x \geq -23$ and $x > -23$ the same?
- b. How are the solutions to $x \geq -23$ and $x > -23$ different?
- c. Graph the solutions to each inequality on the number lines.

$x \geq -23$



$x > -23$



3. Complete the following.
- a. With your partner, determine which inequality best represents each context.

Nutritional Information		
	Calories:	2,000
Total Fat	Less than	65g
Sat Fat	Less than	20g
Cholesterol	Less than	300mg
Sodium	Less than	2,400mg

Total Fat

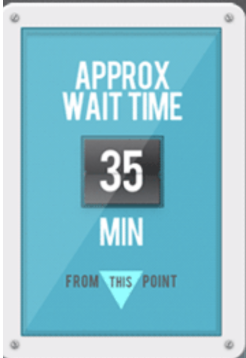
$f \leq 65$ $f \geq 65$

$f < 65$ $f > 65$

YOU MUST BE AT LEAST **54"** TALL TO RIDE

$h \leq 54$ $h \geq 54$

$h < 54$ $h > 54$

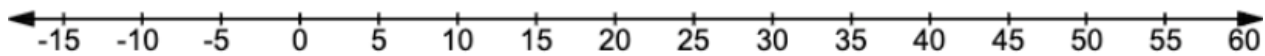
 $p \leq 10$ $p \geq 10$ $p < 10$ $p > 10$	 $s \leq 50$ $s \geq 50$ $s < 50$ $s > 50$
 $x \leq 50$ $x \geq 50$ $x < 50$ $x > 50$	 $w \leq 35$ $w \geq 35$ $w < 35$ $w > 35$

- b. In which, if any, of these situations does a value of 0 make sense given the context?
- c. In which, if any, of these situations does a negative number make sense given the context?
- d. In which, if any, of these situations does 1,000 make sense given the context?

11.7.3 Guided Practice: Translating a Written Description into an Algebraic Inequality

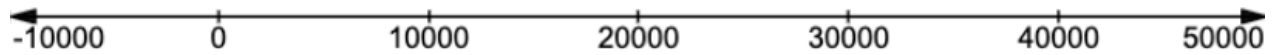
1. According to the Interstate Highway Standards, U.S. and state highway traffic lanes must be at least 15 feet wide.
 - a. Write an inequality to represent the width, w , that traffic lanes can be.

- b. Represent the inequality on the number line.



- c. List two examples of possible widths, in feet, for the traffic lane.
 - d. Give one example of a width that is true for the inequality but does not make sense in the real world.
2. High-altitude parachute jumps are usually made for an altitude that is no more than 35,000 feet to avoid frostbite from extreme cold.
 - a. If h represents the height of a jumper, write an inequality to describe all possible values of h .

- b. Represent the inequality on the number line.

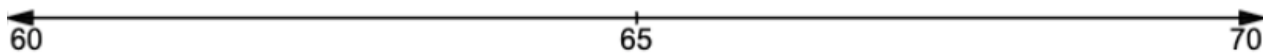


- c. Does the inequality that represents the situation include 0 as a possible height?
- d. Explain why a negative number does not make sense for this context.

11.7.4 Try It: Writing an Algebraic Inequality and Representing It on a Number Line

1. The inequality $67.85 \leq m$ represents the amount of money, m , Edin needs to earn to buy a new bike.

- a. Graph the inequality.



- b. Give one value that is true for $67.85 \leq m$ but is unreasonable for the context.

- c. Explain why the value you chose is unreasonable.

2. A farmer plans to plant more than 52.75 acres of oranges.

- a. If A represents the number of acres of oranges the farmer plants, write an inequality that represents the number of acres the farmer plants.

- b. Circle all the possible values for A , the number of acres the farmer plans to plant, from the list below.

52.7 acres

$52\frac{3}{4}$ acres

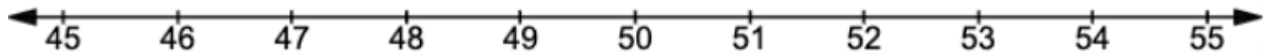
53 acres

52 acres

52.8 acres

$52\frac{1}{4}$ acres

- c. Graph the inequality.



11.7.5 Your Turn!

1. Would you rather drive a go-cart with speed, g , in miles per hour (mph), represented by the inequality $12.5 \geq g$, or an electric bike with speed, b , in mph, represented by $b \geq 7\frac{1}{2}$?

a. I choose the

- ☐ go-cart
 - ☐ electric bike

because _____

_____.

The

- ☐ go-cart
 - ☐ electric bike

could have a speed of 18 mph.

- b. Circle all the values from the list below that are possible speeds for the go-cart.

1.25 mph

15.2 mph

$10\frac{1}{3}$ mph

$\frac{12}{5}$ mph

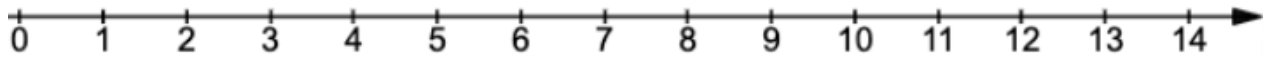
12.45 mph

11.95 mph

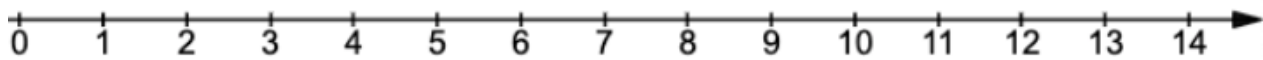
$12\frac{2}{3}$ mph

$11\frac{11}{12}$ mph

- c. Represent the speed of the electric bike on the number line.



- d. Represent the speed of the go-cart on the number line.



- e. Explain why the number lines do not include negative numbers.

11.7.6 Wrap-Up: Ticket Out the Door

At Universal Studios, a person must be at least 48 inches (in.) tall to ride Revenge of the Mummy™.

1. Write an inequality representing the height, h , a person must be to go on the ride.
2. Circle all values from the list below that satisfy this inequality.

47.9 in.

100 in.

 $\frac{48}{2}$ in.

48.5 in.

 $70\frac{3}{4}$ in. $48\frac{1}{4}$ in.

0 in.

400 in.

48.099 in.

48 in.

 $50\frac{3}{4}$ in.

47.9 in.

3. Graph the inequality on the number line.

